Lab 6: bench check before data collection

Use this checklist to select a workable illumination range and video workflow before the full test matrix.

1. Position and log

Radiometer upright; fixed camera; stable lamp; meter beside the rotor plane. Keep background out of the strongest beam. Start with a conservative lamp distance (about 30 cm if compatible with the lamp); adjust only within the lamp's clearance and under instructor supervision. Record lamp setting, distance and lux.

2. Check camera and timing

Save actual capture fps, exposure and available record duration. Record a short preview. Confirm the vanes are distinguishable and a selected event can be followed through one full turn. Write p = ___ events/revolution. Use continuous known-rate phone video if a slow rotation needs more time than Chronos memory allows.

3. Decide feasibility before taking all cases

Try a 60–90 s reference run from a documented initial state. Target at least six consecutive event intervals. If too few occur: improve safe illumination, extend the common window, or reduce the matrix. Do not use playback fps as acquisition fps or create a speed curve from unresolved events.

4. Complete one analysis at the bench

Mark passages → Run_Lab6_Radiometer.m → inspect raw frame events against interval rpm. Record illumination-on frame. Verify the unit and event-count conversion with one manual interval calculation.

5. Select the class matrix

Choose A, B and C from measured lux levels that give usable movies. Core runs: A1, B1, C1, then cooled repeat A2. Use a common time window and record actual covered times. If time permits, add light-off and a labeled warm-return test. Check whether early and late speeds differ before describing a final speed as steady.

What the equipment supports

Supported: turn count, rpm, direction by observation, lux/distance comparison, early/late changes and thermal history. Not measured: internal pressure, vane temperatures, exact Kn, absolute torque or efficiency. The required notebook activity illustrates a limiting collision mechanism; it is not a calibrated model of this bulb.